ATM — the next generation

P L Clarke, N J P Cooper and D J Sutherland

Current ATM service offerings use permanent connections established at subscription time to support user requirements. This is satisfactory given the present number of users. However, in the future, there will be many more users and applications that require infrequent connections at high data rates. To support these, ATM connections-on-demand are required.

This paper considers the way in which a switched ATM service can be supported, and describes the experiences of providing a European trial of such a service within a collaborative project.

1. Introduction

This paper introduces the switching and multiplexing technology, asynchronous transfer mode (ATM), and describes how such a service must support connections-on-demand in order to evolve.

The paper then outlines the progress made in establishing a switched ATM service in Europe by the JAMES project.

2. Overview of ATM

ATM is a connection-oriented communications technology that requires data to be split up into fixed-length cells before being transmitted into a network. Cells are 53 bytes in length, 48 octets allocated to data, with a 5-byte header. The use of fixed, small cell sizes allows for very fast switching times, low transport delays and the ability to mix different types of services (data, voice, video) on a single link, and in a single network.

In an ATM network, points are interconnected by virtual paths, each of which may contain a number of virtual circuits. The cells carry the routeing information in the form of path and circuit identifiers, i.e. virtual path/circuit identifier (VPI/VCI) values. This allows for many different data streams to share the same connection. A good introduction to ATM technology is provided by Adams [1].

2.1 Switched virtual circuits (SVCs)

Initial ATM offerings (for example, BT's CellStream service) uses permanent virtual paths or channels (PVPs/PVCs) to transport traffic. When customers request an ATM connection, they specify their requirements in terms of end-points and service characteristics, at subscription time. Connections are established on their behalf by the service supplier, and that connection is permanently in place across the network. This service is adequate for customers who require a limited connectivity, but becomes unmanageable as the number of end-points that have to be interconnected rises. The number of PVCs required for a full inter-connectivity of N end-points is given by the formula $N(N\text{-}1)/2 \cong N2/2$. Thus it is not sensible to support large numbers of customers requiring full interconnectivity using PVCs. Some form of switching is required.

Future ATM offerings will have the capability to set up connections on demand, use them as required and then clear them down. This is the essence of an SVC capability. It will be of benefit to network users as they only need to pay for a connection as they need it, and will also benefit the network operator who no longer has to maintain permanent bandwidth allocations across the network, which typically will often not be carrying any traffic.

2.2 ATM signalling for SVCs

In order to support connection establishment on demand, customers must have a way of signalling their requirements to the network. ATM connections, however, are much more difficult to describe and support than telephone connections, as the number of options is much higher:

- there are a variety of types of ATM class of service (CBR, ABR, VBR (constant/available/variable bit rate), etc), with different characteristics, each requiring a different level of bandwidth to be reserved by the network to support the call,

- customers generate bursty traffic and it becomes difficult to predict how much bandwidth must be allocated to support the connection — the network must use a connection admission control (CAC) algorithm to decide whether it can support any particular call,

- users must signal their requirements for peak and average traffic rates,

- the network must police a connection to ensure users do not exceed their specified bandwidth allocation.

ATM signalling protocols must be defined both at the interface between the customers and the network, the user/network interface (UNI), and at the interface between the switches, the network/network interface (NNI), to enable ATM connections to be established on demand. In addition there must be a numbering scheme in place to enable the users to identify the required destination.

Both the ATM Forum (ATMF) and the International Telecommunications Union (ITU-T) have been active in developing signalling protocols in these areas.

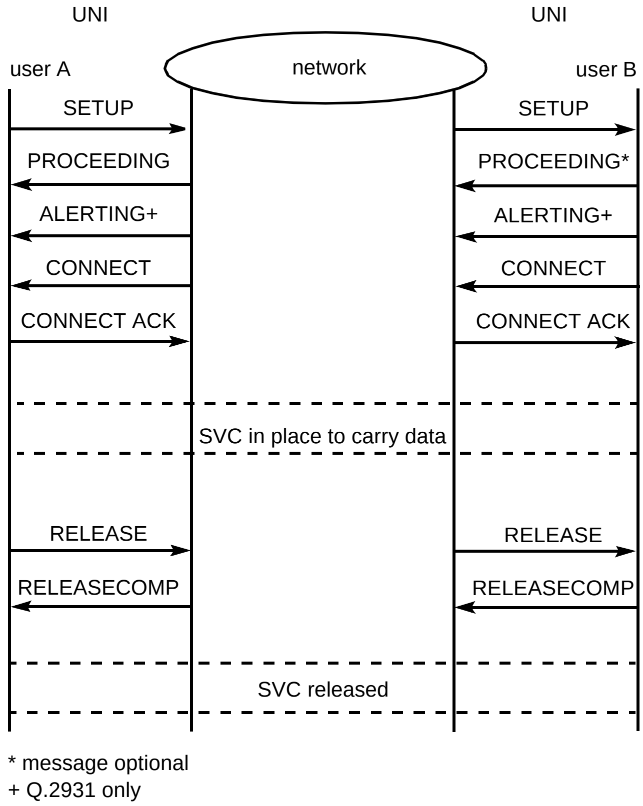

Fig 1 Signalling messages to control an ATM connection.

3. User/network signalling

Users must use standard messages to signal to the network their desire to establish a call, and for this they must use some sort of protocol.

The defined ITU-T UNI protocol is Recommendation Q.2931 [2, 3], which evolved from the narrowband ISDN signalling message set. The ATMF has used this as the basis for its UNI recommendations, ATMF UNI 3.1 [3] (and the upgrade to version 4.0 signalling). An earlier ATMF protocol UNI 3.0 will not interwork with UNI 3.1. Current ATM switches typically support the ATMF UNI 3.1 and 3.0 protocols.

Q.2931 and UNI 3.1 define a number of message types and the sequence in which they must be exchanged between customer and switch and vice versa to establish and subsequently clear down a connection. Each message type contains a number of information elements which specify the details of the call to be established. A typical exchange of messages is shown in Fig 1. A call is initiated using a SETUP message, which contains a call reference value to uniquely identify this call attempt. All subsequent messages passed over this interface associated with this call will carry the same value of call reference. In addition the SETUP message must contain information elements to specify:

- traffic parameters (the ATM traffic descriptor),
- broadband bearer capability,
- quality of service information,

and a number of optional information elements may be used to further specify the details of the call. For further explanation of these elements, the reader is referred to Q.2931 or ATMF UNI 3.1 [3].

The network acknowledges the SETUP with a CALL PROCEEDING message, and (using NNI signalling between the nodes) generates a SETUP message towards the recipient of the call to determine if the called party will participate.

The Q.2931 protocol allows an (optional) alerting message that may be sent. This is useful if the decision on acceptance of the call must be made by a human, and there is a delay while the destination terminal ‘rings’.

The CONNECT message is sent by the called terminal to the network to accept the call, and by the network to the calling party to indicate the call is successful. CONNECT messages must be acknowledged by a CONNECT ACKNOWLEDGEMENT, and an SVC is now established across the network over which the parties can exchange information. This SVC is identified to the calling and called parties by the connection identifier information element which is present either in a SETUP or one of the subsequent messages.

Calls are released by an exchange of RELEASE and RELEASE COMPLETE messages across the UNI interfaces of all parties involved in the call. Either side can initiate a release.

Other message types are also defined by the UNI protocols. These include a STATUS ENQUIRY used to elicit information from a signalling peer, or a NOTIFY message sending information relevant to the call.

SVCs are not necessarily point to point. The UNI protocol supports the establishment of point-to-multipoint connections which might be used for broadcast or multicast transmissions.

Signalling messages must be transmitted using ATM cells, and a virtual circuit with values of VCI 5, and a default VPI of 0 is reserved on each ATM interface to support UNI signalling. Signalling may be either associated or non-associated. In the first case the user requests a virtual channel in the virtual path connection carrying the signalling VC. In the latter case the signalling entity may also allocate VCs in VPs other than that carrying its signalling.

4. Network/network signalling

As with the UNI interface, a number of different signalling protocols are currently defined to operate at the network/network interface (NNI). Both the ITU-T and the ATMF have been active in this area, but unlike the UNI this has resulted in a number of differing protocols.

4.1 B-ISUP

This is the network signalling protocol defined by the ITU-T and is developed in a series of specifications (Q.276x) [4—8]. The protocol is a development of the narrowband NNI protocol. It assumes the customers attached to an ATM network will be identified by E.164 numbers, and it specifies both the messages that must pass between network nodes and also the mapping between the B-ISUP messages and the Q.2931 UNI messages and their respective information elements.

4.2 Private network/network interface(PNNI)

This is the ATMF specification for NNI to be used in private networks and it specifies both the signalling messages required to support SVCs and also a routeing protocol which enables switches to exchange routeing information and to dynamically build routeing tables. The first version of this specification was issued in March 1996, with version 2 currently in progress. Specifications are available from the ATMF website [9].

4.3 B-ICI v2.0

This is the ATMF intercarrier interface which allows different network operators to work together [9]. It has been based on B-ISUP.

4.4 Interim interswitch signalling protocol (IISP)

This is an ATMF recommendation and is almost the same as the UNI 3.1 signalling protocol.

The majority of the present generation of switches are based on the ATMF specifications and support IISP protocols at the NNI. There is now a move, however, to supplement switch functionality by also including the routeing capabilities of the PNNI protocol.

5. Addressing

In order to support an SVC service, there must also be an addressing scheme in operation to enable customers to identify each other. There are two broad approaches to addressing, that of the ITU-T which specifies E.164 addresses, and the ISO which defines a twenty octet network layer service access point (NSAP) address for private networks. The ATMF recommends use of the ISO format for addresses of private networks and refers to NSAPs as AESAs. The following address formats are now available.

- E.164 — the format of addresses used in public telephone and N-ISDN networks

  For ATM addressing, the international format will be used and the numbers can be up to 15 digits (i.e. 8 octets as numbers are encoded using BCD, that is two digits are carried in each octet). The E.164 is padded with as many leading 0s as are needed to make the length of 15 octets and the single semi-octet of binary 1111 is added at the end.

- ISO addresses

  These are NSAP addresses constructed using either the international code designator (ICD), data country code (DCC) or E.164 NSAP. In the E.164 case this provides for the encoding of an E.164 number within a 20 octet NSAP format. Support of this is optional within the ITU-T.

ITU-T-based ATM switches need only support the E.164 addresses, as support of E.164 NSAP is optional within the ITU-T. Most switches on the market, however, use the ATMF protocols and these use E.164 NSAP, DCC or ICD address formats. Given this multiplicity of standards, an ATM service provider will typically have to provide some means of interworking between customers using the different address formats.

6. CellStream, an example of a PVC-based ATM service

ATM is now an established technology, and BT launched an ATM service, CellStream, in March 1997 [10]. Currently CellStream offers permanent virtual paths or permanent virtual circuits on a point-to-point basis. These are provided with a guaranteed quality of service and throughput.

CellStream is planned to support an SVC capability in the future.

7. The JAMES project

JAMES (the Joint ATM Experiment on European Services) is a project comprising 19 European public network operators (PNOs) (see Table 1) which was formally started in April 1996 to build on the work of an earlier European ATM pilot trial [11]. The project was initiated by the JAMES consortium in response to two European Commission initiatives:

- joint Telematics/Esprit Initiative for the validation of advanced services for European research and university networks using bandwidths of 34 and 155 Mbit/s,
- a call under ACTS 1994-1998 framework programme for the interworking of national host facilities.

Table 1 Participants in JAMES

Belgacom	PTT Telecom Netherlands
BT	P&T Luxembourg
Deutsche Telekom	Swiss Telecom PTT
FINNET International	Telecom Eireann
France Telecom	Telecom Finland
Israel (Bezeq)	Telecom Italia
OTE	TeleDanmark
Portugal Telecom	Telefonica
Post & Telecom Iceland	Telenor
Post & Telekom Austria	Telia

JAMES tests and evaluates the provision of new ATM-based broadband services and applications throughout Europe with the objective of:

- establishing user requirements — service type, features, quality of service (QoS),
- prioritising user requirements and defining a trial plan for each service,
- assessing the market potential for trans-European broadband services,
- validating end-user services, especially the use of SVCs,
- obtaining feedback from the users.

7.1 *Project structure*

The JAMES project is organised around three boards (see Fig 2). The Project Board is chaired by France Telecom and addresses the project management aspects. It comprises representatives from each PNO. The Users Board is chaired by BT and addresses marketing, communications and user aspects. Its objective is to manage the relations of projects with the users and as part of this task it arranges regular meetings with JAMES users — the JAMES users forum. Again the Users Board has representatives from each of the JAMES PNOs. The Operations Board is again chaired by BT with representatives from each PNO. It addresses operations and development issues.

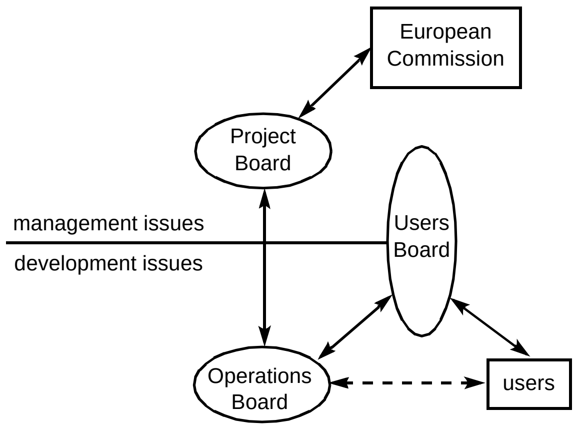

Fig 2 JAMES project hierarchy.

The Operations Board oversees a number of work packages, the activities of which include:

- planning and implementing the network,
- ATM network planning,
- user service descriptions,
- VP bearer network,
- customer support,
- ATM bearer capability,
- security on the network.

The JAMES project offers a number of experimental services to the JAMES users that allow the participating PNOs the opportunity of obtaining experience of supplying and managing broadband and data services over an ATM backbone. As the services are experimental, they are:

- not part of any commercial service (i.e. free to JAMES users),
- provisioned on a piecemeal basis,
- not guaranteed in their availability or reliability.

The experimental services offered are:

- ATM VP bearer services (CBR, VBR, etc) [1],
- SMDS over ATM,
- LAN interconnection service,
- IP over ATM,
- ATM SVC service.

7.2 The JAMES network

The JAMES network consists of ATM switches in each of the participating countries, connected by high-speed links, typically 34 Mbit/s, but with some connections at 155 Mbit/s. Within each country there are typically connections to one or more of the following networks:

- national host networks,
- national experimental ATM networks,
- commercial broadband networks,
- operators' research laboratories.

The current topology of the JAMES network is shown in Fig 3. The different network operators typically use different switches, including Alcatel, FORE, General DataComm, Newbridge, etc.

This network is used to support all the experimental services supported by the JAMES project.

8. JAMES SVC trials

The JAMES network, using a variety of switches from different vendors dispersed throughout Europe offers an ideal opportunity to investigate the problems of offering and supporting an SVC service from the points of view of:

- developing experience of configuring SVCs on switches and the use of appropriate test equipment,
- developing administrative procedures to provision and cease a service to a customer,
- exploring UNI and NNI protocols,
- investigating the problems of fault finding,

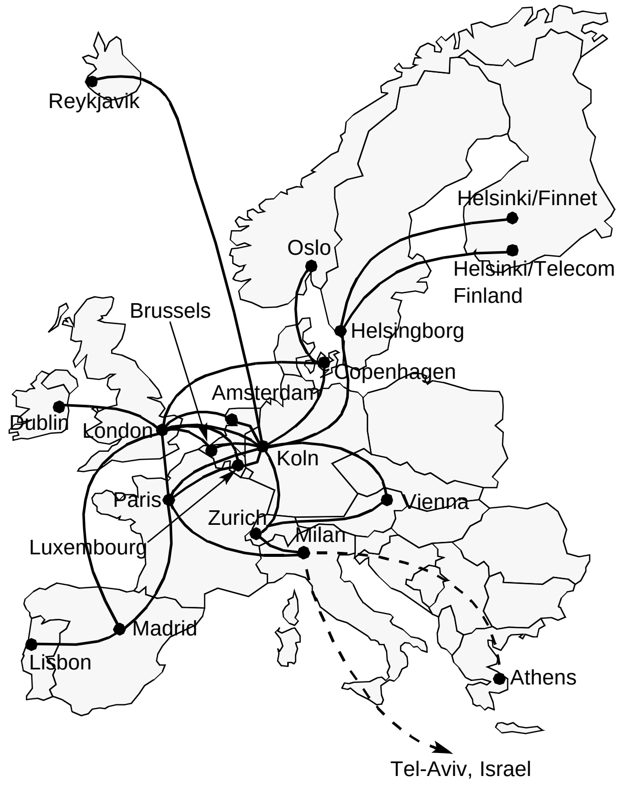

Fig 3 JAMES network.

- investigating the issues of interworking with non-homogeneous switches,
- trials of SVC operation,
- determining the characteristics of SVC traffic.

Until quite recently switch manufacturers either did not support a connection-on-demand function on their equipment, or they implemented proprietary protocols. With the release of a standardised SVC capability on ATM switches, network operators had to become familiar with this new feature and had to prove the reliability and usefulness of the service before they could offer the capability to their customers.

The JAMES approach to implementing a Europe-wide SVC service reflected this need, with an SVC capability being introduced step by step in an evolutionary rather than revolutionary way.

8.1 Test specifications

Initial work was directed towards the development of agreed test specifications. Four levels of testing were identified:

- validation tests — with the aim of building confidence that the switches correctly implemented the UNI and NNI signalling protocols to support an SVC service,

- interoperability tests — to confirm that the different switches interwork with each other, and that the SVC service functioned correctly from the service providers' point of view,

- end-to-end tests — to confirm both that SVCs could be set up between customers using the JAMES network, and that the SVC service functioned correctly from the customers' point of view,

- application tests — to determine the behaviour of standard applications used by customers when supported on an SVC ATM network.

8.2 Selection of interfaces and addressing formats

Before any form of interworking could be attempted, the JAMES partners had to agree on both the NNI protocol to be implemented through the JAMES network and also on the formats of addresses to allow customers to be identified.

The choice of UNI and NNI protocols is determined to a large extent by the capabilities of equipment on the market, which typically adopted the ATMF recommendations. The UNI was selected to be ATMF UNI 3.1. This protocol was also chosen to be implemented at the NNI. The majority of switches support neither the ITU-T protocol (B-ISUP), nor the later ATMF protocol (PNNI). Hence the choice for NNI is either UNI 3.1, or IISP, which is itself a reworking of UNI 3.1.

The choice of addressing for the JAMES network is somewhat more problematical. Although the use of a pure E.164 number is likely to limit the service to be offered, as the first generation of customer equipment needing to use SVCs are likely to use NSAP addresses and possibly to construct the end-system identifier part of the NSAP address field using their media access control (MAC) address, nevertheless:

- some switches used by JAMES partners do not currently support NSAP addresses,

- one JAMES partner currently offers a national ATM service based on E.164 addressing.

Technically a way can be found to develop a common addressing format as it is possible to convert between 20-octet NSAP addresses and the 15-digit E.164 addresses, using the sub-address field to carry the additional NSAP number. However, within the time-scales of JAMES it has not yet proved possible to agree on a common addressing format. Thus JAMES recommends the use of NSAP E.164 addresses where possible, but notes that this is not mandatory.

8.3 Collaborative interworking

The end of 1996 saw the completion and agreement of the various test specifications within the JAMES project, with a significant number of partners having obtained switches and testers to progress the work. The time-table agreed for 1997 follows:

- partners to complete a local validation of their switch capabilities using an agreed minimum compliance test list by May 1997,

- JAMES partners, having completed this milestone, are then able to move on to interoperability and end-to-end tests in collaboration with another partner — the time-scale for completion of this bilateral testing phase is the end of July 1997,

- following a successful bilateral test, partners to engage in trilateral or multilateral testing to develop a number of 'islands' of SVC capability within the JAMES network — the date for completion of this phase of the work is the end of September 1997,

- the SVC islands to be linked to form a coherent SVC capability before the end of 1997,

- customers can be introduced on to the network at any phase after bilateral testing, and it is hoped the customer base will increase as the SVC overlay increases in size.

The means of implementing the SVC overlay has, to date, been to support SVC functionality at the switches at the periphery of the JAMES network, and to interconnect these switches using permanent virtual path connections (PVPCs). Thus, in the network, SVC traffic is carried by PVPCs. Technically this is not elegant because:

- a PVPC used to support traffic from SVCs will take up bandwidth, even if customers have cleared the SVCs and there is no traffic to carry,

- there is no way to predict or to dynamically allocate the amount of bandwidth needed for a PVPC to support SVCs. This will depend on the number of simultaneous SVC calls and the requirements needed to support each call.

However, for the first phase of bi- and tri-lateral testing, PVPC tunnelling allows JAMES members to:

- interconnect on an *ad hoc* basis without the operators of the transit nodes needing to be familiar with the configuration of an SVC capability,

- allocate numbers to customers on an *ad hoc* basis — there is no need for unified routeing tables within the network,

- reduce the interaction of the SVC trials with the other JAMES experimental services which are based on a PVC network.

As SVC islands are merged to form a global switched capability, it is intended that the interconnection by PVPCs be phased out, allowing the transit nodes to support signalling.

8.4 SVC customers

The users of the JAMES network include the JAMES project itself, National Research Networks, such as the TEN-34 consortium [12], and other EC-funded research projects, including ACTS [13], and TEN-IBC.

The demand for SVC connections by the early JAMES users, however, is only growing rather slowly. Typically JAMES users are running IP applications using CPE that either does not support the protocols required to signal the need for SVC connections, or does not implement traffic shaping. Traffic shaping is necessary to regulate traffic so that it conforms to the characteristics (peak cell rate, cell delay variation, etc) of the established ATM connection. Traffic which is not shaped is liable to be discarded by a policing function within the ATM network.

The JAMES SVC working party is actively encouraging customers to use the developing SVC capabilities of the JAMES network, and a growth in customers is expected as the service migrates beyond the bilateral testing phase.

8.5 Speed to provision the network

Because of the experimental nature of the work of exploring SVCs, a number of factors affect the rate of progress as follows:

- many of the switches used for the provision of the JAMES network are provided by the PNOs collaborating within the project — this has the benefit of guaranteeing a mix of different vendors' equipment, but means delays as each PNO has to allocate kit for the use of JAMES out of their existing budgets,

- as different switch manufacturers have different priorities for the provision of switch capabilities, it takes time for the JAMES collaborators to determine and make available switches for an SVC service,

- as the service is predicated on new switch capabilities, there are delays while operators spend time becoming confident with using the new capabilities of the kit — in addition, the performance of validation tests typically requires the purchase of additional test equipment to understand the reasons for service failures,

- as the switches are located away from the working environment of the research engineers for some PNOs, collaborative tests require a day of travel — this precludes *ad hoc* experimentation,

- as some PNOs have to share the switch-supporting JAMES SVCs with other projects, work on the switch requires the advanced booking of switch resources (again precluding *ad hoc* experimentation), and the tear down of working switch configurations once a test phase has been completed,

- limitations of bandwidth within the core JAMES network means that some PVC tunnels supporting SVC testing have to be torn down at the conclusion of a test phase to release the bandwidth for other JAMES testing.

The speed of provision of the SVC network is therefore slow and the partners typically experience frustration in their attempts to provide permanent islands of SVC capability or indeed a permanent readiness for testing.

9. Conclusions

An ATM service supporting SVCs provides the means for customers to use services requiring high bandwidths in a timely, flexible and cost-effective manner. The protocols to support this functionality have now been proved in a Europe-wide trial.

It is planned that the JAMES project will finish at the end of March 1998. The experimental nature of the SVC work has allowed the project partners to work with newly introduced standards and equipment in a way which would not have been possible on a commercial service. This has been particularly true of the experiences gained as to the interoperability of equipment from different manufacturers. The experiences of the JAMES project have already been fed back to equipment manufacturers.

BT has its own national service, CellStream, which currently offers PVCs. An obvious way in which this service could evolve would be to offer SVCs, and the experiences of the JAMES project would then be put to good use.

Acknowledgements

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13 Advanced Communications Technologies and Services (ACTS) home page, http://www.uk.infowin.org/ACTS/

Peter Clarke graduated from Leeds University with a PhD in Polymer Physics in 1982. He joined BT in 1984 working initially on X.25 services and narrowband ISDN. Subsequently he became involved in studies of broadband services. He developed an implementation of the signalling protocol Q.2931 and worked in a team considering the suitability of DAVIC protocols to offer broadband services. Currently he is supporting BT’s SVC work within the JAMES project.

Nick Cooper joined BT Laboratories in 1973 after graduating from Cambridge University. His work has included X.25 packet switch design and the support of packet switching within the narrowband ISDN environment.

He currently leads a team exploring signalling and call control issues for ATM and represents BT on signalling matters at the ATMF.

David Sutherland joined BT as a sponsored student in 1970. After graduating from Imperial College, London in 1974 he investigated the effects of crosstalk on digital systems using local cables, which led to the introduction of ISDN transmission systems by BT. After a research exchange with NTT in 1988, he has led BT’s involvement in a number of European collaborative projects in the automotive, aerospace and social security sectors. He currently leads the SVC work within the JAMES project.